Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST ANDREW'S JUNIOR COLLEGE
2014 JC2 Preliminary Examinations

H2 BIOLOGY **9648/3**

Paper 3: Applications and Planning Question
(Mark Scheme)

Tuesday 16 September 2014 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,
1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/14
2	/14
3	/12
5	/20
Total	/60
4 (Planning)	/12

This document consists of **??** printed pages.

[Turn over

Answer **all** *questions*.

1. The human erythropoietin gene has been isolated from a DNA library. The gene is contained in a HindIII-BamHI fragment (see **Fig. 1.1**). The gene is 2144 bp long and it encodes a 27-amino acid signal peptide and a 166-amino acid mature protein. Erythropoietin is a hormone (glycoprotein) which plays an important role in the production of circulating red blood cells.

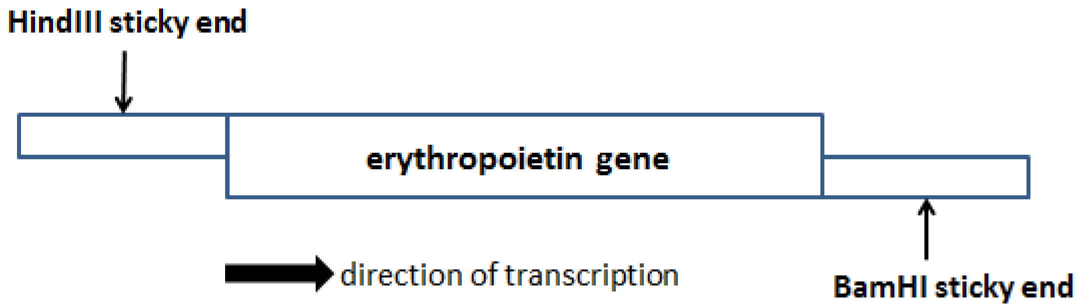


Fig. 1.1

(a) State the type of DNA library where the erythropoietin gene was obtained from.

....[½]

1 Genomic DNA library

A bacterial plasmid (obtained from *Escherichia coli*), pUC19, with multiple cloning site containing PstI, HindIII, BamHI and SmaI is used as a cloning vector. **Py** may be one of two pyrimidine bases and **Pu** may be one of two purine bases. The DNA bases of the enzyme's target site are shown in **Fig. 1.2**. The cleavage sites are shown by means of arrows.

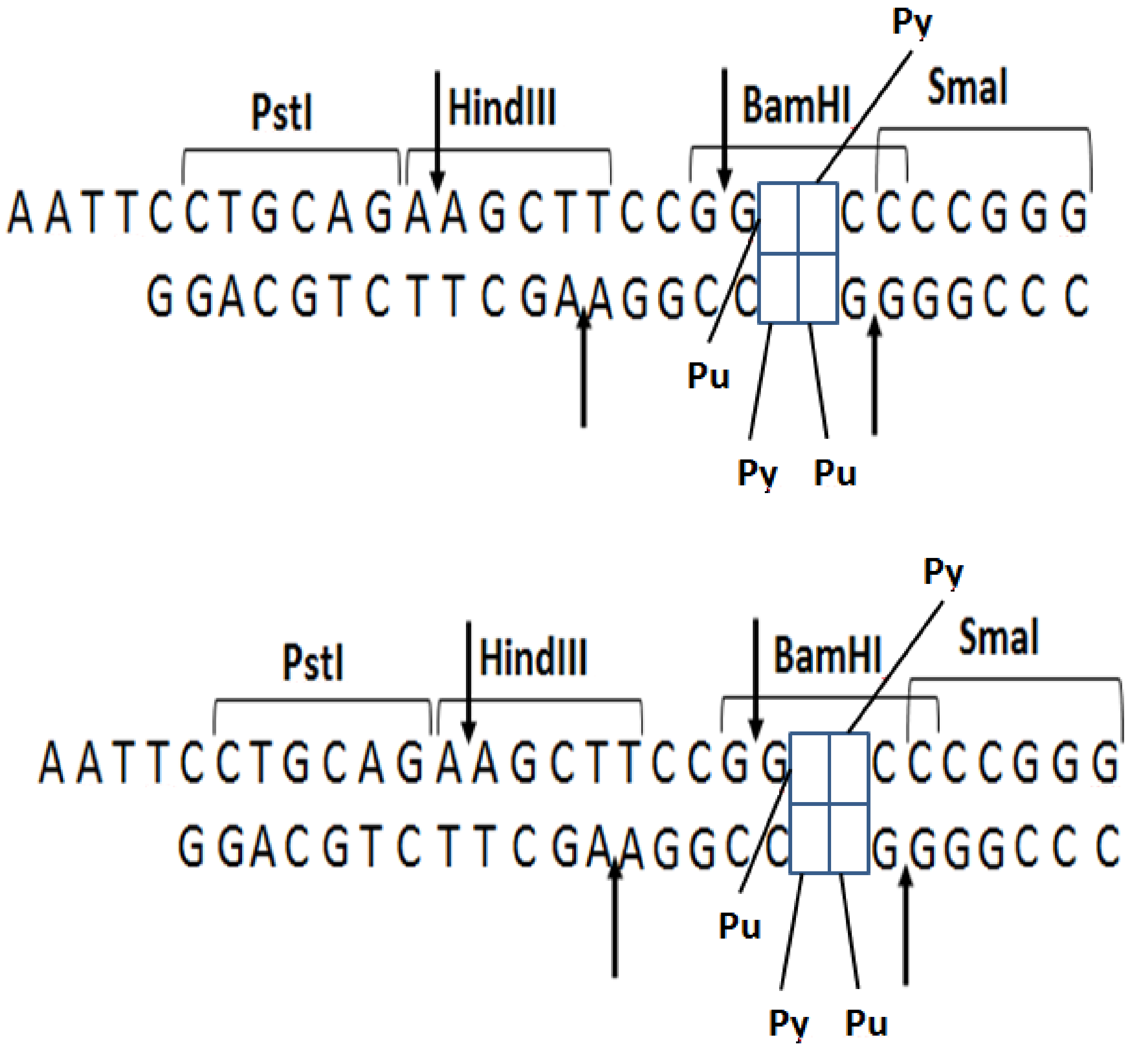


Fig. 1.2

(b) Complete **Fig. 1.2** to show two possible arrangements of bases of the target site of *BamHI.*

…………………………………………………………………………………………………[1]

1 A T
2 G C

(c) Describe the natural function of a restriction enzyme such as *BamHI.*

…………………………………………………………………………………..………………..…..[1]

1 Cleave any <u>foreign DNA</u> that enters the bacterial cell, thus
2 protecting <u>bacteria</u> from attack by bacteriophages / virus.

To form recombinant plasmid (pUC 19 with the inserted erythropoietin gene shown in **Fig 1.1**), pUC 19 is cleaved with *Hin*dIII and *Bam*HI. **Fig. 1.3(b)** shows bacterial plasmid pUC 19 after enzymes cleavage.

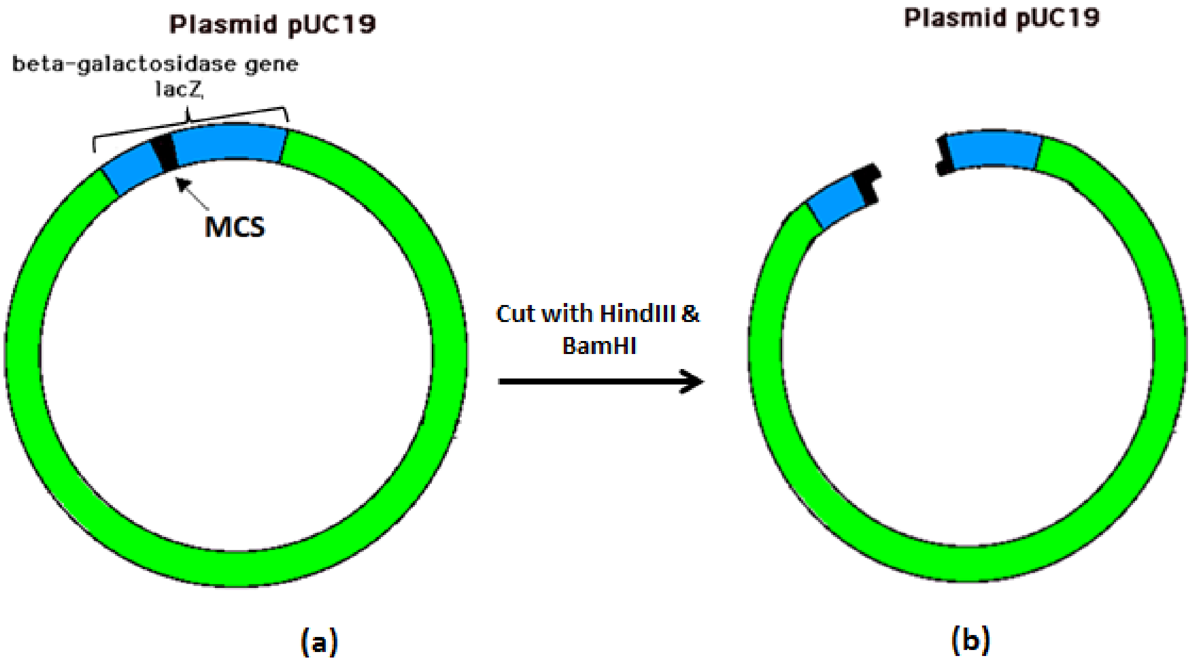
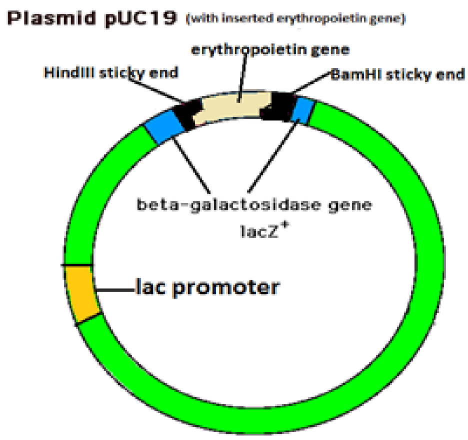


Fig. 1.3

(d) Indicate on the **Fig. 1.3 (b)** the positions of
- lac promoter,
- the inserted erythropoietin gene showing clearly the position of HindIII and BamHI sticky ends.

……………………………………………………………………………………………..[1]



1 Correct position of lac promoter before the inserted erythropoietin gene
2 Correct position of inserted gene with HindIII and BamHI sticky end indicated on the gene (with relative position of lac promoter close to the HindIII sticky end)
REJECT: if gene is not labelled

(e) Suggest one reason why different restriction enzymes were used to create erythropoietin gene with different sticky ends on each end of this gene.
……………………………………………………………………………………………..…..…...[1]

1 allows the gene of interest to be inserted into the plasmid (cut by the same restriction enzymes) **in the correct orientation**
2 downstream of lac promoter for **transcription to occur**

(f) Describe the subsequent steps in the formation of recombinant plasmid.
………………………………………..……………………………….……………...……[2]

1 Use the same restriction enzymes, HindIII and BamHI, to digest DNA and plasmid to generate <u>complementary sticky ends</u>
2 The plasmid and the DNA are **mixed** together
3 the sticky ends will **anneal** by complementary base pairing
4 <u>DNA ligase</u> added to seal the nicks
5 by catalyzing the formation of <u>phosphodiester bonds</u> (between adjacent nucleotides on the sugar phosphate backbone)

(g) Outline the principle of insertional inactivation of the *lacZ* gene.
……………………………………...……………………………….………...……[3]

1 *lacZ* gene codes for <u>functional β-galactosidase</u>;
2 culture medium contains <u>X-gal</u>,
3 a chromogenic substrate (colourless) that when hydrolysed / cleaved by β-galactosidase, will produce a <u>blue</u> product

4 colonies containing non-recombinant plasmids (bacteria containing the plasmid without inserted erythropoietin gene at *lac Z*) will appear blue
5 due to <u>the intact</u> *lacZ* gene (which codes for functional β-galactosidase)

6 colonies containing recombinant plasmids (bacteria containing the plasmid with inserted gene at *lac Z*) will appear white
7 as insertion of erythropoietin gene <u>disrupted the *lacZ* gene</u>
8 resulting in <u>non-functional β-galactosidase</u> produced

(h) The erythropoietin gene, when introduced in the host cells, *Escherichia coli*, produces erythropoietin that is biologically inactive. Explain why.

……………………………………………………………………………………………………[2]

1 Presence of introns
2 No RNA splicing / no spliceosomes in bacterial cells
3 (Introns will be used in translation,) causing additional amino acids to be incorporated into the protein
/ Presence of stop codon in intron leads to premature termination of translation

4 No organelles in bacterial cells for post-translational modification (as erythropoietin is a glycoprotein)

(i) It is possible to create an expression vector to carry a eukaryotic gene that can be used within bacteria and yeast cells. State and explain what components are necessary to be incorporate into such vector.

……………………………………………………………………………………………....[2½]

An expression vector to contain the following components:
1 prokaryotic promoter and eukaryotic promoter (just upstream of the MCS where the eukaryotic gene can be inserted in correct reading frame)
2 allows recognition by host cell's RNA polymerase for expression of inserted gene in bacteria cell and yeast cell

3 origin of replication recognisable by prokaryotic and eukaryotic enzymes
4 enables independent replication of the plasmid (and foreign gene) inside the host cell.
5 resulting in multiple copies of the plasmid and foreign gene within one bacterium to be formed.

[Q1 Total: 14]

2. People with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because the gene encoding for enzyme Adenosine deaminase (ADA) is mutated.

Genetic screening is used to test for the presence of recessive alleles which cause ADA SCID. Cell samples are obtained from mouth swaps or from the blood.

A RFLP marker linked to the ADA gene from one of a pair of homologous chromosomes is shown in **Fig. 2.1**. The target sites of *EcoRI* are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

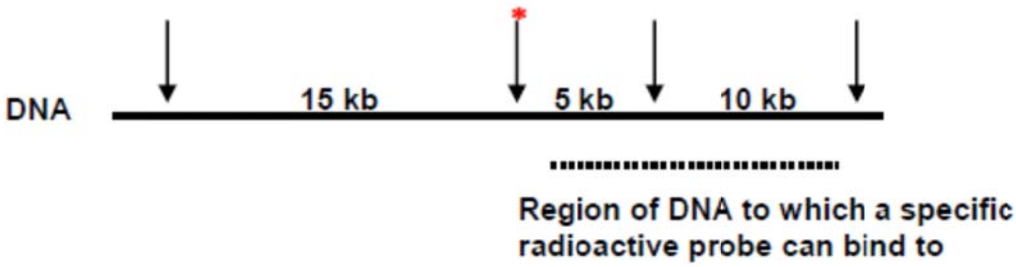


Fig. 2.1

A mutation alters one base of the coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other who is heterozygous for this mutation is cut with *EcoRI* and the DNA fragments separated by gel electrophoresis.

(a) On **Fig. 2.2** below,

(i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individuals tested and label the size of the fragments produced on the gel.
…..………………………………………………………………………………………….[2]

(ii) Circle the fragments which will combine with the radioactive probe and appear as black bands on an autoradiography.
…..………………………………………………………………………………………….[1]

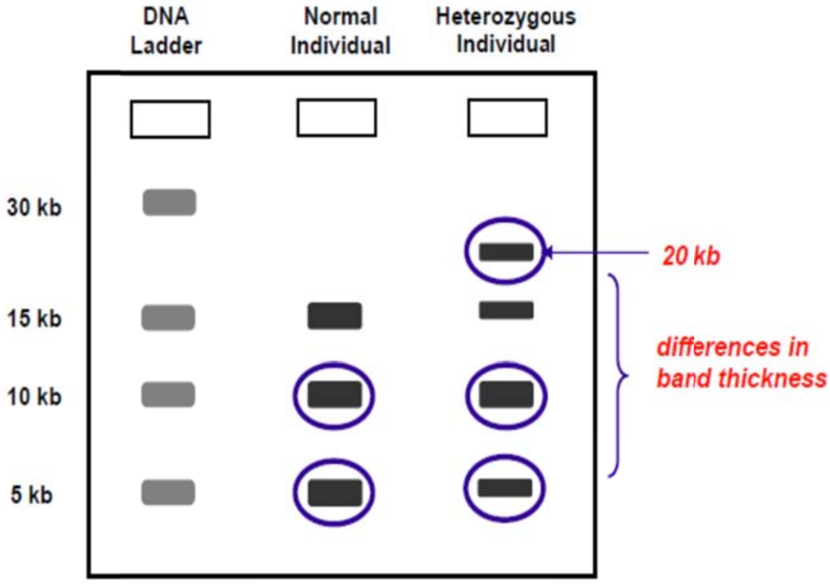


Figure 2.2

(i)

1 Normal individual – 3 correct bands
2 Heterozygous individual – 4 correct bands
3 Difference in band thickness
4 Correct labeling for drawn bands

(ii)

5 Circling of 10kb and 5kb bands for normal individual
6 Circling of 30kb, 10kb and 5 kb bands for heterozygous individual

(iii) Suggest why cells from mouth swabs or blood samples are used rather than gametes.

…………………………………………………………………………………………………..[1]

1 The mouth swabs / blood samples will contain diploid cells / 23 pairs of chromosomes / have two set of chromosomes;
2 as compared to the gametes which are haploid / 23 chromosomes only / have one set of chromosomes;

(b) Explain how the lack of a functional gene coding for ADA leads to SCID.
………………………………………………………………………………………………[3]

1 <u>deficiency</u> of **functional** enzyme <u>ADA</u>;
2 which catalyzes the <u>deamination</u> of adenosine and deoxyadenosine;
3 <u>adenosine and deoxyadenosine</u> **accumulation**;
4 <u>toxic</u> to lymphocytes;

5 <u>block replication</u> of lymphocytes / <u>prevent formation</u> of lymphocytes;
6 leaving insufficient <u>T and B lymphocytes</u> to fight off infections (compromised immune system);

7 susceptible to opportunistic infections /
e.g. pneumonia / meningitis / chronic diarrhea;
/ children usually die (if not treated)

(c) Explain why a person without SCID can have the diseased RFLP marker.
………………………………………………………………………………………………[1]

1 (During gamete formation), crossing over occurred at loci <u>between the RFLP marker and ADA gene</u>;
2 caused normal ADA allele to be linked to diseased RFLP marker;

Some patients with SCID have been treated with blood stem cells.
(d)(i) Explain how the properties of blood stem cells allow them to be effectively used to fight SCID.
………………………………………………………………………………………………[3]

1 blood stem cells are <u>undifferentiated / unspecialized</u> cells;
2 <u>multipotent;</u>
3 **able to differentiate** into specialized blood cell types / give rise to a limited range of cell type / produce only cells of a specific lineage
4 e.g. <u>T and B lymphocytes</u> to replace disease cells lacking ADA in SCID;

5 capable of <u>self-renewal</u> / dividing rapidly by mitosis;
6 ref. asymmetric division to generate continuous supply of terminally differentiated cells for long-term treatment;

(ii) Suggest why the *ex vivo* method was used in this case.
………………………………………………………………………………………………[1]

1 stem cells can be easily extracted and **isolated**
2 ref. easily **cultured** in the laboratory;
3 ref. easily transfected
4 allow for **selection** of successfully transformed cells;

REJECT:
- Prevent the retrovirus from targeting other rapidly proliferating cells in the body;
- viral vector can regain ability to cause disease

(e) Gene therapy using the same blood stem cells and retrovirus vectors has succeeded for 15 French baby boys suffering from ADA SCID. However 3 of the little boys later developed cancer and died. Explain how this is possible.

…..…..……………………………………………………………………………………..[2]

1 insertional mutagenesis / most transgenes insert ectopically;
2 ref. activation of oncogenes;
3 inactivation of (2 copies) of the tumor suppressor gene;
4 resulting in loss of arrest of cell division to allow more time for the cell to repair its DNA + loss of ability for DNA repair + loss of apoptosis;
5 ref. uncontrolled cell division/proliferation;

[Q2 Total: 14]

3. Environmental stress severely restricts the distribution and productivity of crop plants. Some species of crop plant produce a substance called glycinebetaine (GB) which is a molecule known to protect plants against stress.

Scientists transferred the gene for GB into a species of crop plant that does not normally produce GB. The plants were then cloned by tissue culture to produce genetically modified plants.

(a) Suggest one advantage of cloning the plants by tissue culture.

……………………………………………………………..………………………........…[1]

1 genetically identical
2 possess the desirable features of the stock plants

3 allows the rapid multiplication of plants
4 giving rise to bulk production of genetically identical plants

5 new plants are disease-free;
6 leading to higher yield / better quality

7 genetic modifications possible (with help of protoplast cultures)
8 desirable traits can be introduced

9 Take up little space when compared with plants growing in fields
10 can be grown intensively

11 Plantlets are light and small in size
12 can be air-freighted and transported easily / cheaply / in large quantities, increasing international trade

13 Independent of climate changes so plants can be produced continuously/at any time of year
14 flexibility in meeting consumer demand

15 possible to standardise the conditions for growth and obtain many batches of identical plants
16 ensures product uniformity

These genetically modified plants then produced GB. The scientists grew large numbers of the same crop plant with and without the gene at different temperatures. After 3 days, they found the increase in dry mass of the plants.

Figure 3.1 shows their results.

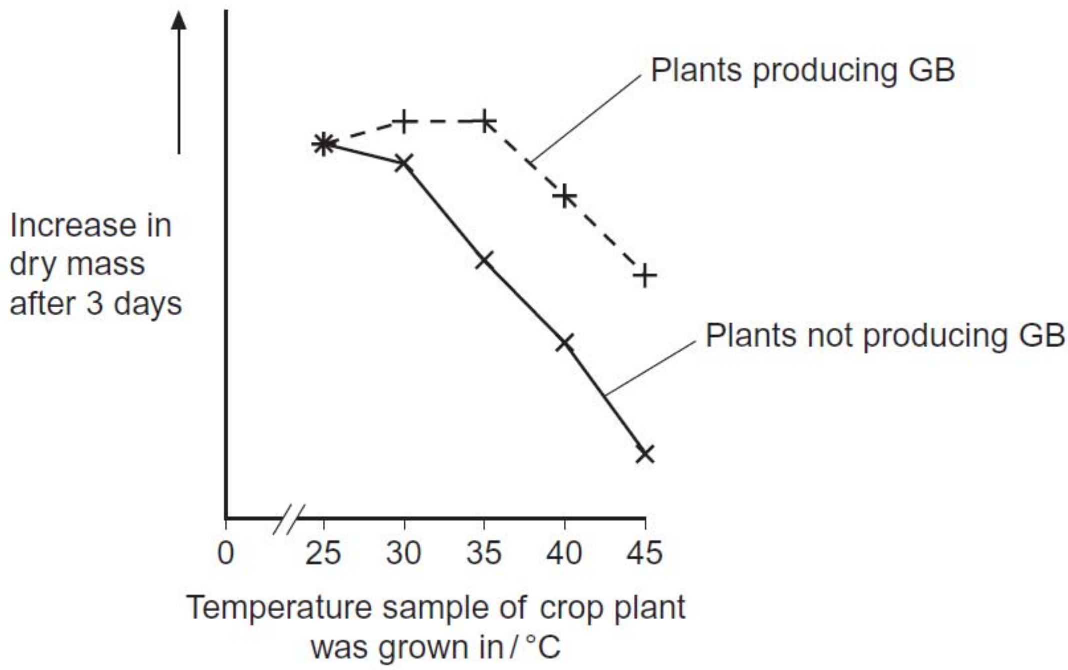


Fig.3.1

(b) Describe the effect on growth of transferring the gene for GB into this plant.

……………………………………………………………………………………………………[2]

1 Increase in dry mass is correlated with positive growth ;

2 No effect at 25^0C ;
3 Continuous growth at 30^0C and 35^0C ;
4 Above 35^0C, increase in dry mass drops ;

5 Ref. Grows more than plant without gene for GB.

The scientists measured the rate of photosynthesis in plants that produce GB and plants that do not produce GB at 25 ºC, 35 ºC and 45 ºC.

Fig.3.2 shows their results.

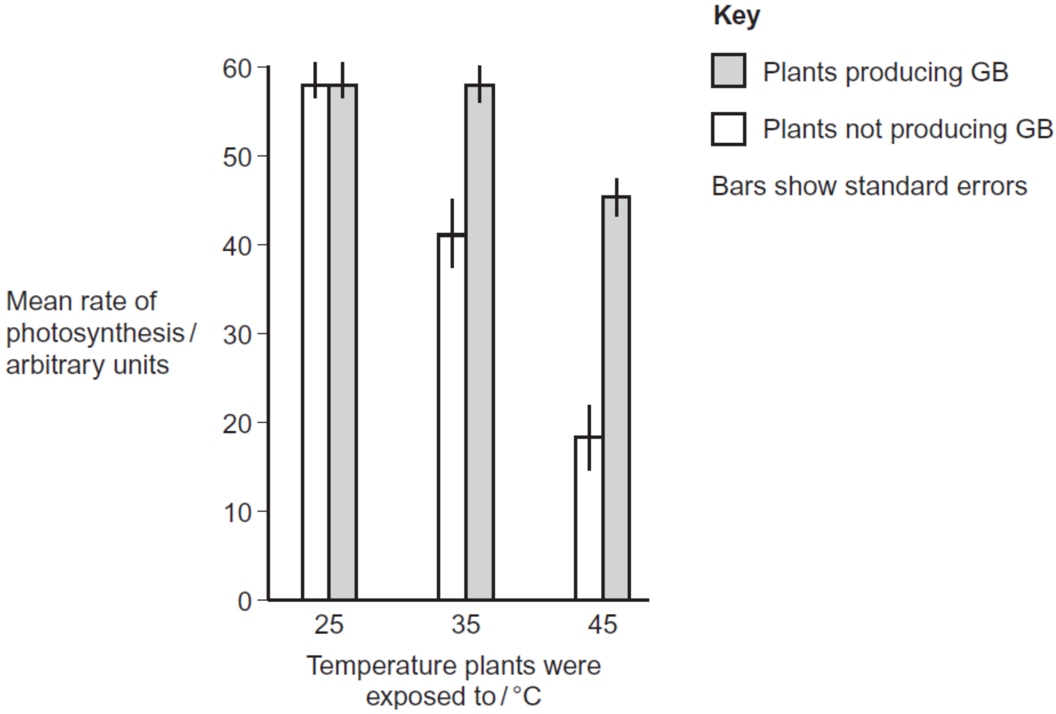


Fig.3.2

(c) The scientists concluded that the production of GB protects photosynthesis from damage by high temperatures. Use these data to support this conclusion.

……………………………………………………………..…………………………...….[2½]

1 At 25^0C, **no difference** in mean rate of photosynthesis between GB plants and non-GB plants;
2 Both at 58 a.u. (accept 57 a.u.)

3 At 35^0C, plants producing GB recorded a **higher** mean rate of photosynthesis than non-GP producing plants ;
4 58 a.u. (accept 57 a.u.) compared to 41 a.u (accept 42 a.u.)

5 At 45^0C, plants producing GB recorded a **higher** mean rate of photosynthesis than non-GP producing plants ;
6 45 a.u. (accept 46 a.u.) compared to 18 a.u. (accept 17 a.u.) ;

(d) Use the data from **Fig.3.2** for plants that do no produce GB to explain the effect of temperature on changes in dry mass of the plants shown in **Fig.3.1**.

……………………………………………………………..…………………………........…[3]

Effect

1 At higher temperatures , rate of photosynthesis is lower ;
/ at 25^{0}C, 35^{0}C and 45^{0}C, rate of photosynthesis is 58 a.u. 41 a.u. and 18 a.u. respectively;

2 as dry mass is lower;

Explanation

3 Enzyme activity reduced / enzymes denatured ;

4 Decreased carbon fixation;

5 resulting in less sugars formed ;

6 Less organic subtances made that contributed to dry mass ;

7 Lower rate of respiration ;

8 Less ATP produced;

9 for named function associated with growth e.g. mitosis, uptake of mineral ions;

A transgenic animal is one that carries a foreign gene that has been deliberately inserted into its genome. The foreign gene is constructed using recombinant DNA methodology. In addition to the gene itself, the DNA usually includes other sequences to enable it to be expressed correctly by the cells of the host.

Atlantic salmon (foreground) which normally grows in Spring and Summer was genetically modified to produce the Aquadvantage® salmon (background).

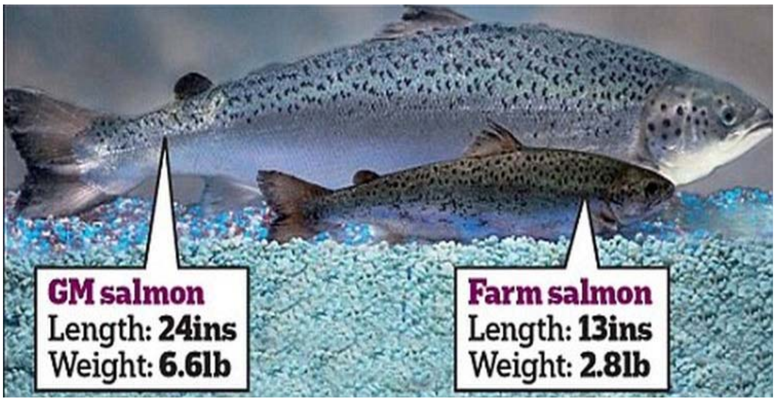


Fig.3.1
Credits : http://foreverconscious.com/wp-content/uploads/2014/04/gmo-salmon-compare.png

(e) Explain why the genetically engineered Aquadvantage® salmon (GM salmon) is considered a transgenic animal.

……[1½]

1 Active growth hormone gene from Pacific Chinook salmon ;
2 Combined with regulatory sequences / promoter of the ocean pout ;
3 Inserted into genome of fertilized Atlantic salmon eggs ;

(f) Describe the effect of the genetic modifications carried out on the GM salmon.

……[1]

1 GM salmon produces higher levels of fish growth hormone ;
2 Accelerated growth rate of fish ;
/ Reaches its desired market length in a shorter period of time ;
REJECT GM salmon grows to larger size

(g) Explain the significance of the transgenic GM salmon in solving the demand for food in the world.

……[1]

1 More fish can be harvested in a year ;
2 Allows salmon to grow all year around (instead of only during Spring and Summer).

[Q3 Total: 12]

4. Planning question

Germination in seeds is a complex physiological process triggered by imbibition of water. As water enters the seeds, their enzymes become activated. Activated enzymes start respiration and use organic matter stored in the seed during respiration to make ATP molecules needed for growth.

You are required to plan an investigation into the effect of temperature on the rate of respiration of germinating seeds.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- Seeds that have been soaked for 24 hours
- Glass beads
- Soda lime
- Syringes
- Glass capillary tubes with bore diameter of 0.4 mm
- Rubber connecting tubings
- Beaker of coloured liquid
- Ruler marked in mm
- Paper towels
- Stop watch
- Electronic weighing balance accurate to 0.1 g
- Incubator

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- describe the method with scientific reasoning so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results table and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Q4 Total: 12]

1. Theoretical consideration or rationale of the plan to justify the practical procedure (2 marks)

(2 pts – ½ mark, 3 pts - 1 mark, 4 pts – 1 ½ marks, 5-7 pts – 2 marks)

a) O_2 is used as the final electron acceptor in the ETC. The protons and electrons recombine with oxygen to form water.
b) CO_2 is produced during oxidative decarboxylation during the link reaction and the Krebs cycle

c) Higher rate of O_2 uptake indicates a higher rate of respiration. At optimum temperature, rate of respiration is highest.
d) As temperature increases, kinetic energy increases, more ES complexes formed, rate of respiration increases.
e) When temperatures are too high, 3D structure of enzymes involved in respiration disrupted, enzymes are denatured, rate of respiration decreases.

f) Soda lime absorbs any CO_2 produced by the germinating seeds. The pressure in the vessel decreases due to the uptake of O_2 during respiration and air is sucked from the capillary to keep the pressure constant.
g) The oxygen uptake is detected by the displacement of the drop of coloured liquid.

2. Independent variable (1 mark)

a) Temperature
b) At least 5 temperatures, regular intervals (10-50°C) <u>using incubator</u>

3. Dependent variable (1 mark)

a) Rate of respiration = rate of volume of O_2 uptake / $mm^3\ min^{-1}$
b) Volume of O_2 uptake = $\pi r^2 \times d$
r = internal radius of capillary tube
d = distance moved by coloured liquid in a specific time

4. Variables to be kept constant (1 mark)

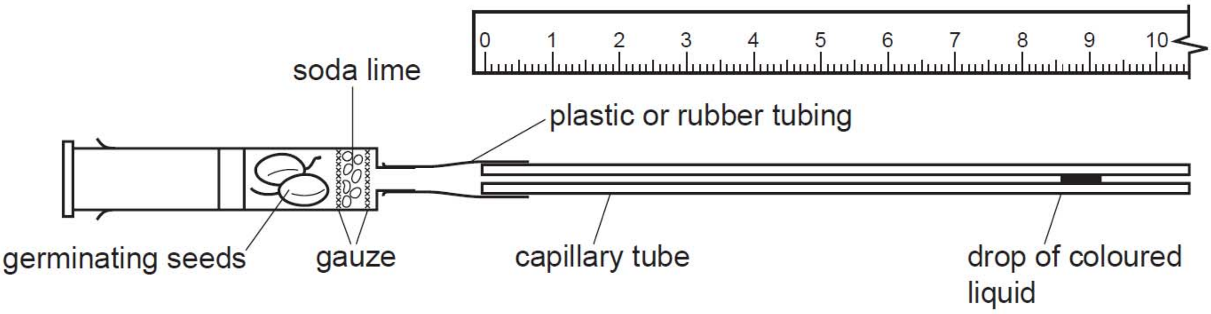
a) <u>Specified mass</u> of germinating seeds, using weighing balance (Reject number of seeds)
b) <u>Specified mass</u> of soda lime, using weighing balance
c) Duration of measurements, using stop watch

5. Controls and justification (1 mark)

a) Replace germinating seeds with equal mass of glass beads, same conditions as experimental setup
b) <u>Purpose</u> of control - To show that oxygen uptake is due to the germinating seeds undergoing respiration and not other factors
c) Setting up of a <u>compensation tube</u> and use results to compensate for natural movement of fluid (due to change in air volume at different temperatures)

6. Procedures (2 marks)

a) Description of respirometer setup
b) Labelled diagram
c) Capillary action to introduce drop of coloured liquid into tube
d) Specified time for equilibration (Accept: 1-10 min)
e) Specified time for measurement of distance moved by coloured liquid using ruler (Accept: 5-10 min)



Remove the plunger from the syringe and place soda lime inside the syringe. Weigh 5 g of the germinating seeds and place them in the syringe barrel and replace the plunger by pushing it in until it is about 0.5 cm from the germinating seeds. Connect the glass capillary tube securely to the syringe using a rubber connecting tubing.

Dip the end of the glass capillary tube into the coloured liquid (i.e. manometer fluid) so that a drop enter the capillary tube. Remove any excess liquid with paper towelling.
Place the respirometer horizontally inside the incubator shelf.
Place a ruler beside the capillary tube.

Wait for 3 minutes to ensure equilibration of temperature between the respirometer and the incubator. Also ensure that the drop of coloured liquid is moving smoothly towards the syringe.
Record the distance travelled by the drop of coloured liquid in 5 min.

7. Reliability (1 mark)

a) Remove plunger to replace air / oxygen between measurement (to prevent respiration becoming anaerobic)
b) Respirometer must be airtight / adjust position of meniscus with plunger ;
c) 3 replicates for each temperature and 3 repeats using fresh germinating seeds and new set-up

8. Results table with appropriate headings (1 mark)

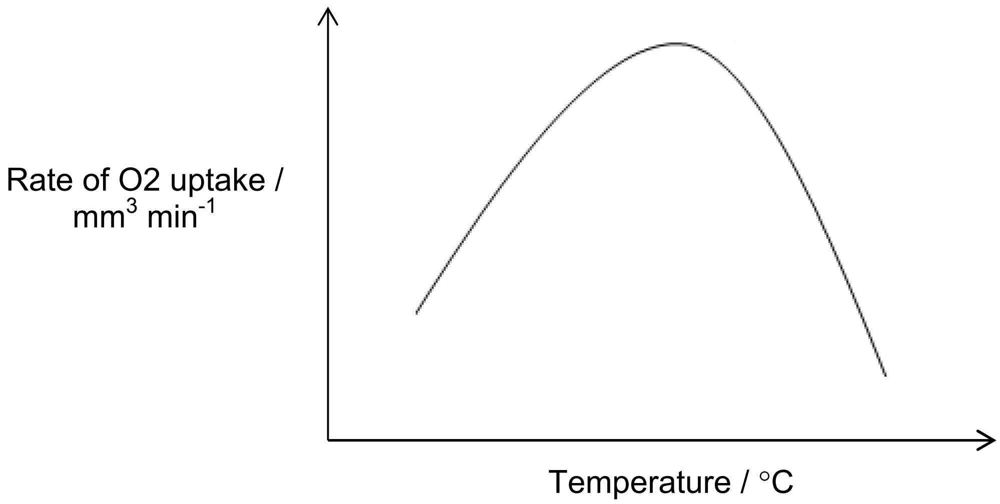
½ mark ½ mark

Temperature / °C	Distance travelled by coloured liquid in 5 minutes / mm				Volume of O_2 uptake / mm^3	Rate of O_2 uptake / mm^3 min^{-1}
	Reading 1	Reading 2	Reading 3	Average		
10						
20						
30						
40						
50						
60						

9. Theoretical graph with correct labels and units (1 mark)

a) Respiration rate (Rate of O_2 uptake / mm^3 min^{-1}) against temperature (°C)

b) Draw a theoretical graph



10. Risk and safety measures (1 mark)

a) Avoid contact with soda lime as it is corrosive; wear rubber gloves when handling it

b) Glass capillary tube is fragile and sharp when broken; handle and dispose broken pieces carefully

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism. [6]

(b) Discuss the goals and benefits of the human genome project. [7]

(c) Discuss the ethical concerns that have arisen from the human genome project. [7]

[Total: 20]

(a) Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism. [6]

Definition of RFLP

1 There is genetic variation between individuals
2 The difference in nucleotide sequence is due to mutation
3 Restriction enzymes only cut DNA at specific restriction sites.
4 RFLP refers to differences in the restriction sites on homologous chromosomes that leads to different restriction fragment patterns.

How to detect RFLP (written in general context)

5 restriction digestion of genomic DNA produces numerous fragments /smear on the gel
6 radioactive probe is used to distinguish the restriction fragments of interest from among many other restriction fragments

Process of nucleic acid hybridisation

7 Restriction enzymes are used, to cut DNA strands into fragments
8 Agarose gel electrophoresis is performed, to separate the DNA fragments based on size
9 The gel is treated with sodium hydroxide to cause the double-stranded DNA to denature / separate into single strands.
10 DNA fragments are transferred onto nitrocellulose membrane
11 then baked to permanently crosslink the DNA to the membrane.
12 membrane is treated with a single-stranded
13 radioactive DNA probe which is complementary to the target sequence / marker / restriction fragment of interest
14 after hybridisation, excess probe is washed off from the membrane
15 pattern of hybridization is visualized on X-ray film / description of X-ray autoradiography

(b) Discuss the goals and benefits of the human genome project. [7]

Goals (max 3)

1 Identify all the genes in human DNA (ref. 30,000 genes)
2 Determine the DNA sequences (ref. 3 billion base pairs)
3 Store this information in databases and make the sequence totally and freely accessible
4 Improve tools for data analysis
5 Transfer related technologies to the private sector
6 Address the ethical, legal, and social issues (ELSI) that may arise from the project.

Benefits (max 4)

Genetic testing

1 improve diagnosis of disease (e.g. detecting rare alleles involved in common disease)

2 genetic testing or screening
/ detect genetic predispositions to disease e.g. allow the identification of gene variants that are important for the maintenance of health, particularly in the presence of known environmental risk factors

Pharmacogenetics (max 1)

3 facilitates the study of complex diseases involving multiple genes and their interactions with other genes, and environmental factors
4 create drugs based on molecular information
5 design “custom drugs” (pharmacogenomics) based on individual genetic profiles
6 use of gene therapy to treat certain genetic diseases

Comparative genomics

7 HGP data allows for comparative genomics / comparative studies with genomes of other model organisms. This is useful for development of animal models of disease development.
8 Allows quick identification of genes encoding proteins found to be involved in cancer/disease through homology searches of identified gene sequences with database sequences.

DNA Identification (max 1)

9 identify potential suspects whose DNA may match evidence left at crime scenes
10 exonerate persons wrongly accused of crimes
11 identify crime and catastrophe victims
12 establish paternity and other family relationship
13 match organ donors with recipients in transplant programs

Bioarchaeology, Anthropology, Evolution, and Human Migration (max 1)

14 Study evolution through germline mutations in lineages
15 Study migration of different population groups based on maternal inheritance
16 Study mutations on the Y chromosome to trace lineage and migration of males

(c) Discuss the ethical concerns that have arisen from the human genome project. [7]

[max 1m per idea]

1) Privacy / confidentiality of genetic information
- Difficult to determine who owns and controls the genetic information or who should have access to it

2) Fairness in the use of genetic information in non-health care settings, including the potential for misuse.
- Difficult to determine who should have access to genetic information, eg. insurers, employers, courts, schools, adoption agencies, the military
- Forensic DNA databanks. Ensuring that they are used for the purpose for which they were collected and protected from misuse. Also, where the public has also assisted the police by volunteering genetic samples to assist in the investigations of unsolved crimes, ensure that special protections are put in place for the DNA samples and the information generated

3) Patenting of DNA sequences
- Patenting of DNA sequences may limit their accessibility and development into useful products
- Intellectual property issues surrounding access to and use of genetic information by both researchers and the public)

4) Fairness in the accessibility of data/material/technologies
- May lead to inequity between developed and undeveloped countries in access to and use of new genetic information and technologies to improve human health

5) Clinical issues including the education of doctors and other health-service providers, people identified with genetic conditions, and the general public about capabilities, limitations, and social risks and implementation of standards and quality-control measures
- The education of healthcare professionals, policy makers, students, and the public about genetics and the complex issues that result from genomic research
- The integration of new genetic technologies, such as genetic testing, into the practice of clinical medicine.

6) Ethical issues surrounding the design and conduct of genetic research with people, including the process of informed consent.

7) Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.
- Accuracy, reliability, and utility of genetic test (An individual is much more than the sum of their genes: the individual's environment can modify the expression of genetic messages to the body and many factors are not genetic that make an individual who they are)
- Reliability of the genetic tests for pre-natal decision making, like abortion

8) Reproductive issues such as the adequacy of information for parents / individuals and the rights for decision making for complex and potentially controversial procedures

- Inappropriate applications of genetic testing such as for the sole purpose of family balancing (sexing of a fetus for this reason) or its use in paternity testing without the informed consent of all parties involved

9) Possible discrimination for individuals with known genetic defects / prone to genetic diseases in different areas of life

- e.g. genetic discrimination in employment and insurance
- abortion of fetuses with minor disorders
- psychological impact and stigmatization due to an individual's genetic differences

10) The historical, social, psychological impact of genomics-derived data of race, ethnicity, kinship and individual and group identity

- Use of genetic data to define racial groups, or of racial categories to classify biological traits, is prone to misinterpretation
- The social implications for both individuals and society, of discovering genetic contributions related to human traits and behaviors such as stigmatized by the suggestion that alleles associated with what some people perceive as 'negative' physiological or behavioural traits are more frequent in certain populations

11) Conceptual and philosophical implications regard human responsibility and degree of genetic influence on behaviour and hence the party responsible

12) The line between medical disease and enhancement is controversial. Need for setting of boundaries in applications of the genetics technology.